Literature Review

In this part we'll address the previous solutions and research papers conducted to solve the problem of uneven signal distribution, we'll also examine the gaps by each paper we address and make sure our project can overcome those limitations.

Unlike most traditional systems that rely on broad signal dispersion regardless of the different factors affecting each zone, our project is built on those previous systems but with a few improvements to ensure both complete coverage and precise driven signal enhancement that guarantees specialized signal strength for each location.

We have collected relevant studies by applying the following search strings systematically to the Google Scholar database, querying titles and abstracts to identify pertinent articles for our research:

Search 1: **("WiFi repeater" OR "wireless repeater" OR "WiFi range extender") AND ("coverage extension" OR "coverage improvement") AND ("wireless LAN" OR "WiFi")**

Search 2: **("low-frequency Wi-Fi" OR "4G vs 5G" OR "sub-6 GHz") AND ("coverage range" OR "signal penetration" OR "throughput") AND ("advantages" OR "benefits" OR "disadvantages" OR "limitations" OR "trade-offs") AND ("wireless network" OR "Wi-Fi")**

Search 3: **"Distributed Antenna System" AND "WiFi" AND "indoor propagation"**

Search 4: **"Phased array" AND "antenna" AND "beamforming IC" AND "WiFi"**

These strings were applied to ensure a comprehensive review of the literature relevant to each aspect of our project.

"The selection criteria for the papers included in this study were developed and applied in accordance with the guidelines provided in the ATLAS.ti guide on critical literature reviews, ensuring a systematic and transparent approach." **[6].**

Inclusion criteria:

Language: Studies must be written in English.

Peer-Reviewed: Studies should have undergone peer review.

Focus: Studies must exclusively focus on Wi-Fi network coverage, dead zones, signal strength measurement, or heat map visualization.

Length: A minimum of four pages is required.

Publication date: Studies published between 2020, and the present are eligible.

No Replication: Studies should not duplicate the same idea by the same author(s).

Source Types: Both journal articles and papers included in conference proceedings are acceptable.

Exclusion criteria:

Non-English Language: Studies not written in English are excluded.

Irrelevant Focus: Studies focusing on cellular networks, Bluetooth, IoT, or other unrelated wireless technologies unless directly related to Wi-Fi dead zones.

Irrelevant Source Types: Blog posts, magazines, newspapers, or other grey literature without peer review.

Insufficient Detail: Abstract-only papers or papers with insufficient methodological or empirical detail.

Publication Date: Studies published before 2020.

Repetition by Same Authors: Repeated contributions by the same authors in journal articles and conference papers are not eligible.

Background and Motivation:

1.Coverage Improvement of IEEE 802.11n Based Campus Wide Wireless LANs

A paper published in 2018 by Srivastava et al

Investigated the coverage improvement of IEEE 802.11n based WLANs in campus environments [7]. The researchers identified dead zones by measuring Wi-Fi signal. strength using the NetSpot tool and visualizing the results with heat maps.

The study found that signal strength below -80 dBm represents dead zones where users lose connectivity completely. The authors proposed using Particle Swarm Optimization (PSO) to reposition access points, which resulted in a 40% improvement in coverage. They also suggested using Wi-Fi repeaters as an additional solution for areas with weak signals in Fig.2.

Fig.2 Coverage improvement solutions and dead zone heatmap.

2.Repeater Swarm-Assisted Cellular Systems: Interaction Stability and Performance Analysis

A paper published in 2025 by Bai et al

Investigated the use of repeater swarms to improve coverage in cellular massive MIMO systems [8]. The researchers proposed deploying large numbers of low-cost, full-duplex repeaters that amplify and instantaneously retransmit signals with minimal delay. The study addressed two fundamental challenges:

- preventing destructive positive feedback caused by inter-repeater interaction, and
- quantifying the performance improvement considering that repeaters inject noise and may introduce additional interference.

The authors derived a generalized Nyquist stability criterion for the repeater swarm system and provided easy-to-check stability conditions based on inter-repeater channel amplitudes. They also developed an iterative algorithm that jointly optimizes repeater gains, user transmit powers, and receive combining weights to maximize the weighted sum rate while ensuring system stability. Numerical results demonstrated that deploying 40 single-antenna repeaters in a cell of 1000-meter radius can nearly double the sum rate compared to systems without repeaters.

The results also showed that repeaters should be placed close to cell-edge users with strong line-of-sight links to the base station to fully realize their benefits.

The objective of **Fig. 3**: To determine the optimal geographical location for placing a repeater to maximize signal quality improvement in communication networks , taking into account the quality of the repeater itself (the noise level it adds).

- Uplink: A standard repeater performs best when it is located midway between the user and the base station.
- (b)Downlink: In this direction, regardless of its quality, a repeater performs best when it is very close to the user.

To deploy repeaters effectively, they must be placed close to users experiencing poor coverage (especially in the downlink direction). Placing them near the base station offers little to no benefit.

The red line (repeater noise is very high, $\sigma^{2R}/\sigma^{2B} = 10$)
The green line (no noise in the repeater, $\sigma^{2R} = 0$)
The blue line (repeater noise ratio = 1, $\sigma^{2R}/\sigma^{2B} = 1$)

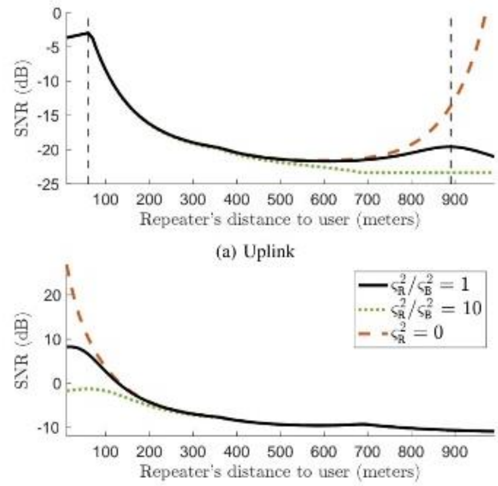

Fig. 3 Optimal repeater placement for uplink and downlink under varying noise ratios.

3.Distributed Antenna Systems (DAS)

DAS is a technology designed to distribute radio signals from a central hub to multiple spatially distributed antennas **"Analysis of Hybrid DAS-Small Cell Architectures for Dense 5G Deployments"** investigates the combination of DAS and Small Cells, in a new system called Hybrid DAS-Small Cell Architecture **[9].** Where Small Cells are compact low-power base stations that enhance the network capacity in specific areas with high user density. The paper focuses on creating an improved network distribution system that overcomes the limitations of each DAS and Small Cells separately.

"Optimal Antenna Placement in a Distributed Antenna System using RSSI Optimization for Indoor-to- Outdoor Propagation" This paper focuses on optimizing the antennas locations to help reduce the path loss at each grid location, further helping in achieving optimum received signal strength indicator (RSSI) **[10].** The approach used is mathematical models to calculate the RSSI at various distances from the antennas by calculating the path loss at each point of the specified are. For instance, in a 12×12 m² room, the RSSI would be as shown in **Fig .4**, if the optimized position is taken into consideration.

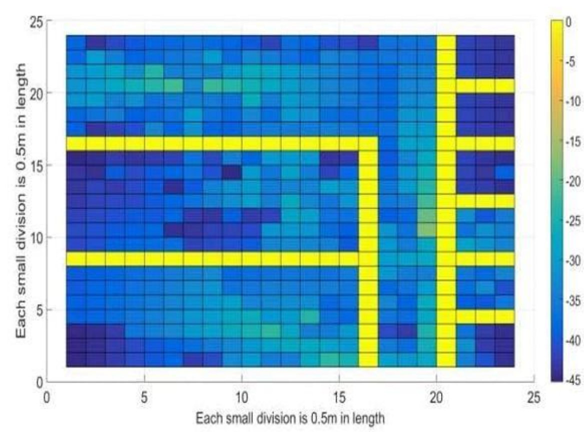

Fig.4 RSSI heatmap

1.Analog IC with Phased Array Antenna

1.1.introduction

Based on the provided sources, here is a detailed research report on **Phased Array Antennas**, their operation, and how we are applying this technology to our innovative signal-boosting project.

What is a Phased Array Antenna? We define a **phased array antenna** as a communication system consisting of multiple radiating elements where the phase of each signal—whether received or transmitted—is individually adjusted [11]. This allows us to steer the radiation pattern (the "beam") in a specific desired direction electronically, without any physical movement of the antenna structure. This technology is essential for modern wireless systems because it increases coverage areas and significantly reduces interference.

Key Uses and Applications In our research, we have identified several critical uses for this technology:

- **5G and Millimeter-Wave (mmW) Communications:** Phased arrays are vital for overcoming the high path loss and low diffraction characteristic of 5G frequencies, which often struggle to reach areas shadowed by buildings.
- **Beamforming:** By concentrating power toward a specific terminal device, we can extend the communication range and improve frequency utilization through spatial multiplexing.
- **Radar and Tracking:** They are used in high-precision radar imaging and target tracking systems.

1.2.Working Principle and Block Diagram Analysis

The core of a phased array's operation is the **progressive phase shift** [12]. By controlling the timing (phase) of the signal at each element, the waves interfere constructively in one direction and destructively in others, effectively "pointing" the signal.

As shown in the **Block Diagram of the proposed phased array antenna** Fig.5 from the sources, the system architecture typically includes the following stages:

1. **Antenna Array:** In this specific design, we use a **four-element slotted waveguide antenna array**.
2. **RF Front-End (Receiving Mode):** The signal from each antenna element enters a dedicated channel. Each channel contains:
   - **LNA (Low Noise Amplifier):** To amplify the weak incoming signal.
   - **Down-Conversion:** The signal is down-converted in stages (e.g., from 9.6 GHz to 2.4 GHz).
   - **Phase Shifters:** These are the most critical components. They adjust the phase of the signal in each channel to maximize the total received power.
3. **Signal Combination:** A **Wilkinson power combiner** merges the four adjusted signals into one.
4. **Demodulation and Control:** The combined signal is sent to an **I/Q demodulator** and then to an **ADC**. A **Power Detector** monitors the signal strength; if the power falls below a certain threshold, a **Controller** unit automatically changes the progressive phase shift to re-steer the beam for better reception.

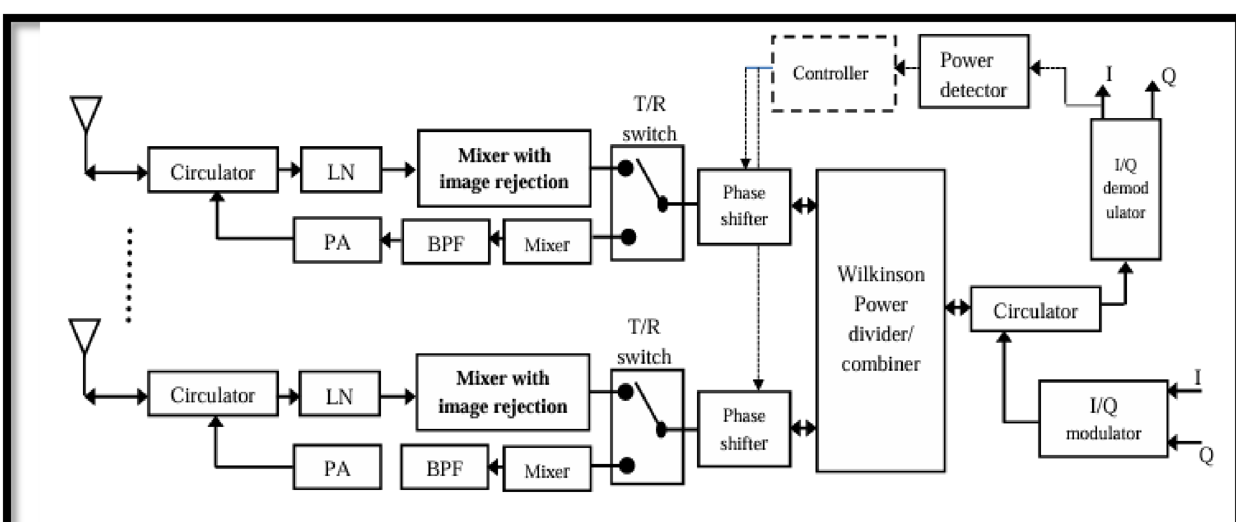

Fig.5 the phased array system

2.reviewing papers and main gaps were concluded:

Solutions	The Gap
Wide Wireless LANs	Temporal signal variations were not considered, as most measurements were taken at fixed times without accounting for daily fluctuations in user density. Interference from coexisting devices operating in the 2.4 GHz band (Bluetooth, ZigBee) was mentioned but not analyzed. The proposed frameworks lack generalizability to different campus architectures and layouts.
Repeater Swarm-Assisted Cellular	The proposed solution was not validated in complex indoor environments such as university campuses or hospitals with numerous obstacles and interfering devices, as the study relied on open-area simulations. It didn't address scenarios with extremely dense repeater deployment, where inter-repeater distances become very small and may cause severe interference despite stability guarantees. The study focused solely on cellular networks and did not validate the approach for Wi-Fi networks, which face different challenges such as wall attenuation and interference from other devices operating in the 2.4 GHz band. Temporal variations in signal strength were not considered, as the research relied on static simulations without accounting for changing user density throughout the day.
Optimization of Wireless Fidelity	The proposed solution was based on manual access point repositioning rather than intelligent optimization algorithms (such as Particle Swarm Optimization), limiting its efficiency and reproducibility. The research did not quantify the improvement with specific metrics (e.g., percentage of coverage improvement or exact latency reduction), relying only on qualitative observations. Temporal variations in user density throughout the day (morning classes vs. night in dormitories) were not fully considered in the optimization strategy.
Distributed antenna systems	The deployment cost of hybrid networks is very high, due to the need for extensive infrastructure, including the fiber optics, base stations, and Small Cells. So it might be needed to give up complete efficiency to meet the financial constraints. Handover delays and packet loss during transitions, which need complex AI-based algorithms and machine learning ML to overcome.
Optimal Antenna Placement	This approach improves the network system distribution, but the system is static, once the antennas are installed at their specified location they can't be physically moved. This static design limits the dynamic adaptability to sudden spikes in traffic demands.

3. Our Project "The Distributed Smart Booster Network"

We are developing a high-precision solution to address localized internet signal degradation. Our system integrates a distributed network of custom-designed Analog ICs with a central phased array antenna to create an intelligent, self-optimizing ecosystem. A **phased array antenna** is a communication system where the phase of signals from multiple radiating elements is individually adjusted to steer the beam electronically, increasing coverage and decreasing interference.

4. System Architecture and Workflow

4.1. system architecture:

Our architecture follows the principles of analog beamforming and signal processing components (such as LNAs and phase shifters) as illustrated in the **Block Diagram of the proposed phased array antenna (Fig.5)**.

4.1. workflow:

- **Distributed Sensor Network (Analog ICs):**
  Instead of a single sensor, we deploy a set of custom Analog ICs strategically distributed throughout the environment. Each IC serves as a localized remote diagnostic node that continuously monitors the "Net" (internet signal) quality at its specific coordinate.

- **Centralized Control via Microcontroller:**
  All distributed ICs and the phased array antenna are interfaced with a single, central Microcontroller. This unit acts as the "brain" of the system, receiving real-time signal telemetry from every sensor node simultaneously.

- **Intelligent Fault Detection and Localization:**
  The Microcontroller processes the incoming data to perform a comparative analysis, identifying exactly which Analog IC is reporting a critically weak signal. Once the weakest nodes are identified, the Microcontroller calculates the required **progressive phase shift** needed to target those specific locations.

- **Dynamic Beam Steering and Signal Boosting:**
  The Microcontroller commands the antenna's **phase shifters** to adjust the signals of the individual radiating elements. This allows the antenna to "lock on" and project a high-gain, concentrated beam directly toward the IC in the weak zone. This electronic steering is vital for overcoming path loss and extending communication range.

- **Access Point Integration and Signal Integrity:**
  Our phased array antenna is physically linked to an access point connected to the primary main router. Because the system uses electronic beamforming to focus electromagnetic energy specifically on the area of need, we can significantly increase the signal strength for the weak zone without altering or degrading the signal strength for other users in the environment.

3. System Advantages

- **Precision Localization:** It can correctly identify the weak points in the whole place.
- **Cost-Efficiency:** Its cost is low as it consists of a distributed sensor network and one integrated phased antenna module.
- **Environmental Versatility:** It can be applicable in any environment, including both indoor and outdoor places.
- **Electronic Agility:** The whole system is dynamic; once the phased antenna is moved anywhere, it can work easily as the beam is steered electronically without physical movement of the antenna structure.
- **Intelligent Optimization:** Our solution overcomes the limitations of previous work by applying intelligent optimization techniques, providing quantitative performance evaluation, and considering variations in user density over time.

4. Conclusion

By combining a distributed sensor array with the electronic agility of a phased array, we ensure that the network dynamically adapts to the users' needs. This "search and track" methodology provides a robust, concentrated internet signal to the most underserved spots in a facility while maintaining overall network efficiency and stability